

DigiLocker Data-Flow Register

Protocol, authentication, encryption, data exchanged & security controls for every communication link — companion sheet to digilocker-architecture-enhanced.svg

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digilocker.yaml

AUTHOR
MODEL DATE
SOURCE MODEL

#	SOURCE	DESTINATION	PROTOCOL	AUTH.	ENCRYPTION	DIRECTION	DATA EXCHANGED	SECURITY CONTROLS
1	C-01 Citizen Web Portal	C-03 API Gateway	HTTPS	Session ID (cookie)	TLS in transit (HTTPS)	Bidirectional	Sent: authentication-tokens. Received: citizen-profile, authentication-tokens.	Encryption in transit (TLS). Authorization: end-user identity propagation. Session-ID scheme not detailed in model — To Be Verified.
2	C-02 Mobile App	C-03 API Gateway	HTTPS	Bearer token	TLS in transit (HTTPS)	Bidirectional	Sent: authentication-tokens. Received: citizen-profile, authentication-tokens.	Encryption in transit (TLS). Authorization: end-user identity propagation. Token scheme (e.g. JWT/OAuth2) not confirmed by model — To Be Verified.
3	C-03 API Gateway	C-04 Authentication Service	HTTPS	Bearer token	TLS in transit (HTTPS)	Bidirectional	Sent: authentication-tokens. Received: authentication-tokens.	Encryption in transit (TLS). Authorization: technical-user (service account). Used for token validation on every gateway request.
4	C-03 API Gateway	C-06 DigiLocker Backend	HTTPS	Bearer token	TLS in transit (HTTPS)	Bidirectional	Sent: citizen-profile, authentication-tokens. Received: citizen-profile, issued-documents, uploaded-documents, consent-records.	Encryption in transit (TLS). Authorization: end-user identity propagation — Backend acts on behalf of the authenticated citizen, not the Gateway's own identity.
5	C-04 Authentication Service	C-05 Aadhaar Identity Provider (UIDAI)	HTTPS	Credentials (technical)	TLS in transit (HTTPS)	Bidirectional	Sent: aadhaar-data. Received: aadhaar-data.	Encryption in transit (TLS). Authorization: technical-user (service account) — pre-registered government API integration. Crosses out of the DigiLocker trust boundary to UIDAI. Aadhaar OTP protocol logic formally modelled in digilocker-auth.spthy.
6	C-04 Authentication Service	C-13 Relational Database	JDBC (encrypted)	Credentials (technical)	Encrypted JDBC connection	Bidirectional	Sent: authentication-tokens. Received: authentication-tokens.	Encrypted JDBC transport. Authorization: technical-user (service account). Destination encrypted at rest (symmetric shared key, per digilocker.yaml).
7	C-06 DigiLocker Backend	C-07 Document Repository	JDBC (encrypted)	Credentials (technical)	Encrypted JDBC connection	Bidirectional	Sent: issued-documents, uploaded-documents. Received: issued-documents, uploaded-documents.	Encrypted JDBC transport. Authorization: technical-user (service account). Destination encrypted at rest (symmetric shared key).
8	C-06 DigiLocker Backend	C-13 Relational Database	JDBC (encrypted)	Credentials (technical)	Encrypted JDBC connection	Bidirectional	Sent: citizen-profile, consent-records. Received: citizen-profile, consent-records.	Encrypted JDBC transport. Authorization: technical-user (service account). Destination encrypted at rest (symmetric shared key).
9	C-06 DigiLocker Backend	C-08 Consent Manager	HTTPS	Bearer token	TLS in transit (HTTPS)	Bidirectional	Sent: consent-records. Received: consent-records.	Encryption in transit (TLS). Authorization: technical-user (service account). Used to check/update citizen consent.
10	C-06 DigiLocker Backend	C-09 Issuer Service	HTTPS	Bearer token	TLS in transit (HTTPS)	One-way (data flows Issuer → Backend on response)	Sent: (none — read-only fetch). Received: issued-documents, digital-signatures.	Encryption in transit (TLS). Authorization: technical-user (service account). Marked read-only in the model. Crosses out of the DigiLocker trust boundary to the Government Issuer.
11	C-06 DigiLocker Backend	C-10 Requester Service	HTTPS	Bearer token	TLS in transit (HTTPS)	One-way (Backend → Requester push)	Sent: issued-documents. Received: (none logged).	Encryption in transit (TLS). Authorization: technical-user (service account). Documents shared only after consent validation via the Consent Manager. Crosses out of the DigiLocker trust boundary to the 3rd-party Requester.
12	C-06 DigiLocker Backend	C-11 Notification Service	HTTPS	Bearer token	TLS in transit (HTTPS)	One-way (Backend → Notification trigger)	Sent: citizen-profile (contact data). Received: (none logged).	Encryption in transit (TLS). Authorization: technical-user (service account). Triggers citizen email/SMS alerts.
13	C-06 DigiLocker Backend	C-12 Monitoring Service	HTTPS	Bearer token	TLS in transit (HTTPS)	One-way (Backend → Monitoring event)	Sent: audit-logs. Received: (none logged).	Encryption in transit (TLS). Authorization: technical-user (service account). Emits audit/security events for investigation.
14	C-08 Consent Manager	C-13 Relational Database	JDBC (encrypted)	Credentials (technical)	Encrypted JDBC connection	Bidirectional	Sent: consent-records. Received: consent-records.	Encrypted JDBC transport. Authorization: technical-user (service account). Destination encrypted at rest (symmetric shared key).
15	C-12 Monitoring Service	C-13 Relational Database	JDBC (encrypted)	Credentials (technical)	Encrypted JDBC connection	Bidirectional	Sent: audit-logs. Received: audit-logs.	Encrypted JDBC transport. Authorization: technical-user (service account). Destination encrypted at rest (symmetric shared key).